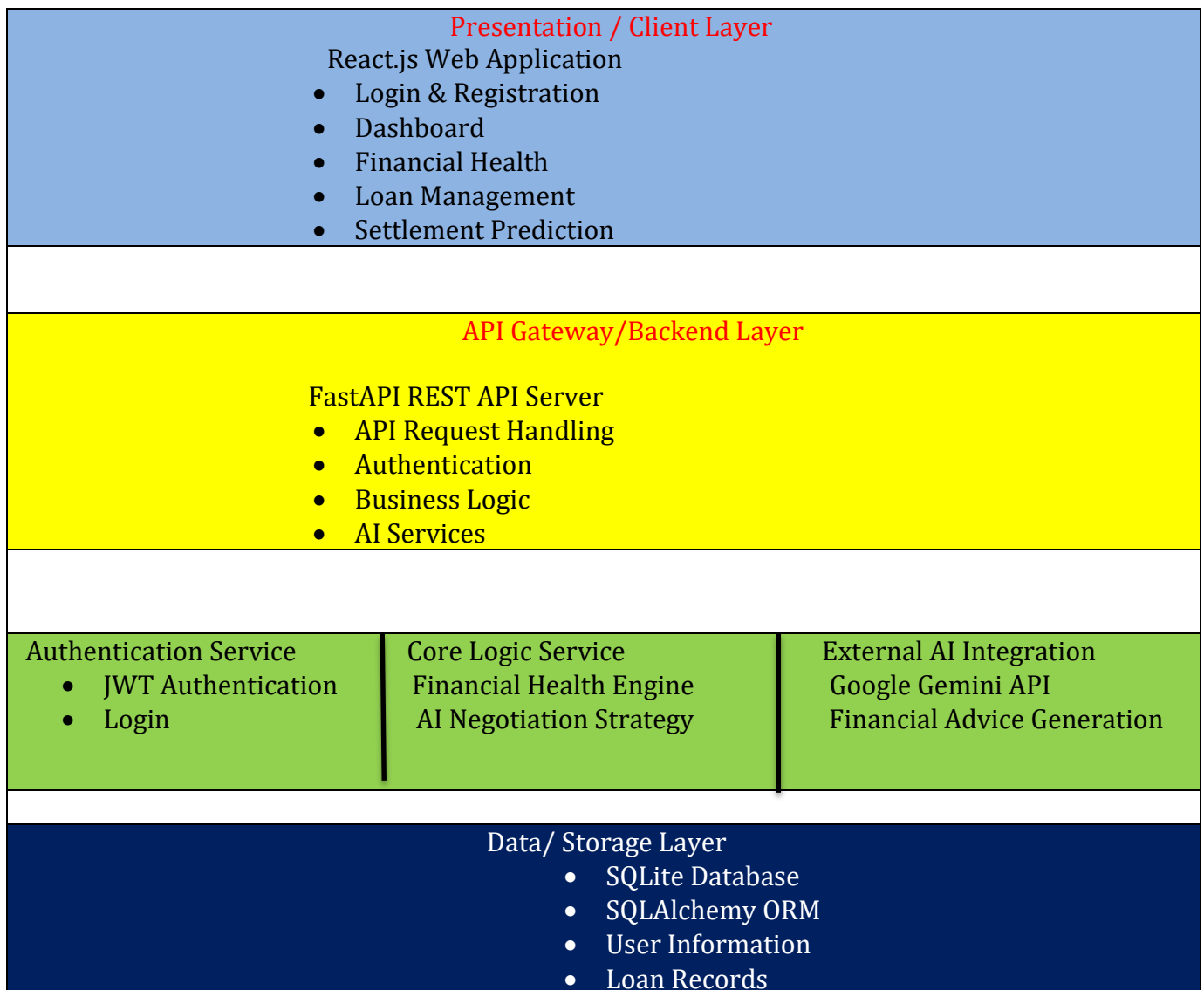


Solution Architecture

Date	2 july 2026
Team ID	XXXXXX
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	1 Mark

Solution Architecture Diagram:



Component Description Table :

Component Name	Description/Role in Architecture	Technologies Used
Presentation Layer	User Interface For login, Dashboard, Loan Management, Settlement Prediction and AI Negotiation	React.js, HTML, CSS, JavaScript
API Gateway	Handles API requests and routes them to backend services	FastAPI, Uvicorn
Authentication Service	User Login, JWT Authentication and Authorization	Python, JWT
Core Logic Service	Financial Health Calculation, Loan Analysis, Settlement Prediction and AI Negotiation Strategy	Python, FastAPI
Database Layer	Stores User Data, Loan Details, AI History and Settlement Records	Python, FastAPI
Database Layer	Stores User Data, Loan Details, AI History and Settlement Records	SQLite, SQLAlchemy ORM